Streamlined Timely Medicine Supply Tracker

PROJECT BASED LEARNING · FULL STACK WEB DEVELOPMENT

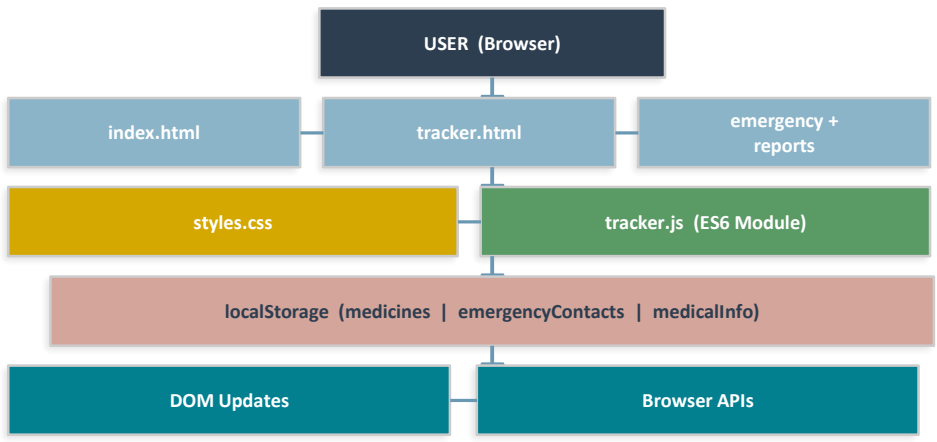
MediTrack — Smart Health Management

Abstract

MediTrack is a full-featured personal medicine management web application. Patients can track daily medications, set snooze reminders, monitor stock levels, and store emergency contacts & allergy information — all within a responsive, mobile-friendly interface.

Built with HTML, CSS, and JavaScript (ES6 Modules), featuring localStorage persistence, real-time DOM updates, async weather integration, CSV export, print functionality, and drug interaction warnings across 4 linked pages.

System Architecture



CO - WISE IMPLEMENTATION

CO1 Internet Fundamentals, HTML & Introductory CSS

- Responsive layout with Flexbox/Grid — 4 linked pages (index, tracker, emergency, reports)
- Semantic HTML5 elements — nav, main, section, header, footer — with ARIA labels for accessibility
- CSS Custom Properties (:root variables) — theme colors, spacing, border-radius, transition speeds
- CSS Animations — @keyframes fadeIn on content load, progress bar transitions, :hover transforms

CO2 Forms, Inputs & Advanced CSS

- Input types: text, tel, number, time, date, checkbox, search with validation attributes
- Form validation — required, minlength, pattern, novalidate HTML attributes
- CSS :focus, :hover, .active pseudo-classes for interactive feedback
- Responsive media queries — mobile nav toggle, grid breakpoints, @media print styles

CO3 JavaScript Fundamentals

- Objects: medicine{} records, INTERACTION_WARNINGS{} dictionary for drug alerts
- Array methods: .filter(), .find(), .map(), .sort(), .forEach(), .reduce()
- Number literals: 0x05 hexadecimal, 6e5 scientific notation for snooze duration
- Template literals for HTML generation; arrow functions; spread operator (...)

CO4 Browser APIs, DOM & Events

- DOM manipulation: createElement(), innerHTML, classList.toggle(), setAttribute()
- Event handling: addEventListener on submit, keyup, storage, window load events
- localStorage: JSON.stringify/parse — medicines, emergencyContacts, medicalInfo keys
- Async APIs: async/await fetchWeather() with Promise; setTimeout for reminders
- setInterval for midnight auto-reset; storage events for real-time cross-page stat sync

CO5 Validation, Error Handling & SEO

- try/catch blocks wrap all localStorage read/write and JSON.parse operations
- Form validation: name ≥2 chars, phone ≥10 digits, stock 0–1000 range checks
- Drug interaction checker — INTERACTION_WARNINGS warns on conflicting medicines
- SEO meta tags: description, keywords, author on all 4 HTML pages
- CSV export via Blob API; auto-download with URL.createObjectURL()

KEY FEATURES

- Medicine Tracker
- Emergency Contacts
- Reports & Analytics
- Snooze Reminders
- Drug Interactions
- CSV Export

Conclusion & Future Scope

This project demonstrates core web development principles — semantic HTML, responsive CSS, ES6 JavaScript modules, async programming, and browser storage APIs — all implemented through a practical, real-world medicine tracking application.

Future Scope: Cloud sync via Firebase, AI-powered drug interaction analysis, push notifications, multi-user authentication, and Progressive Web App (PWA) support for offline access.

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